MT8002 Elements of Set Theory
Course Introduction

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Universidad EAFIT

Semester 2026-2

Outline

Pácto Pedagógico

Introduction

Axiomatic Set Theories

Logical and Mathematical Preliminaries

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Pacto Pedagógico

Como miembros de la Universidad EAFIT, nos comprometemos a actuar de manera íntegra siguiendo los más altos estándares éticos y morales.

- Respeto
- Tolerancia
- Honradez
- Compromiso

Pacto Pedagógico

Página web del curso

`https://asr.github.io/courses/cm0832-set-theory/2026-2/`

Pacto Pedagógico

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Conducto regular, fechas y porcentajes de las evaluaciones

La información está en la página web del curso.

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Responsabilidad compartida

- Profesor
- Estudiantes

Pacto Pedagógico

Asistencia a clase

Reglamento académico de los programas de posgrado, Título II, Capítulo VI, Artículo 62, Parágrafo 2.

> *"El estudiante de posgrado cuyas faltas de asistencia lleguen al treinta por ciento (30%) del total de las horas de clase programadas para el curso o para una parte de éste, cuando se desarrolle con más de un profesor, en secciones temáticas denominadas 'módulo', pierde con calificación de cero punto cero (0.0) del seminario o curso correspondiente y esta nota afecta el promedio crédito acumulado."*

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Orientaciones para el curso

- Se recomienda cuatro horas de trabajo por semana (dos horas por cada hora de clase).
- Las clases son presenciales.
- La evaluación a la docencia es obligatoria.
- Se recomienda revisar periódicamente los canales de comunicación institucionales (EAFIT Interactiva y el correo institucional).
- El estudiante podrá entrar a clase a más tardar 20 minutos después de la hora asignada para su inicio.

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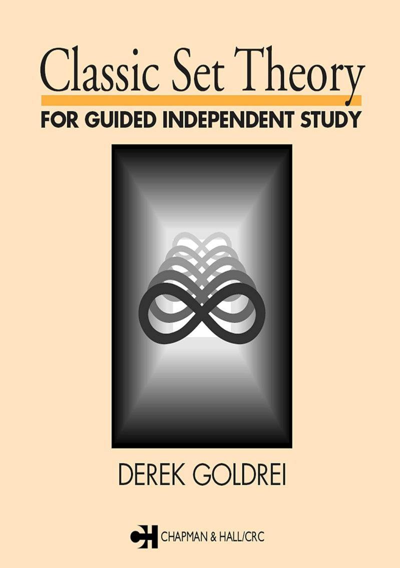

Classic Set Theory
FOR GUIDED INDEPENDENT STUDY
DEREK GOLDREI
CHAPMAN & HALL/CRC

Textbook

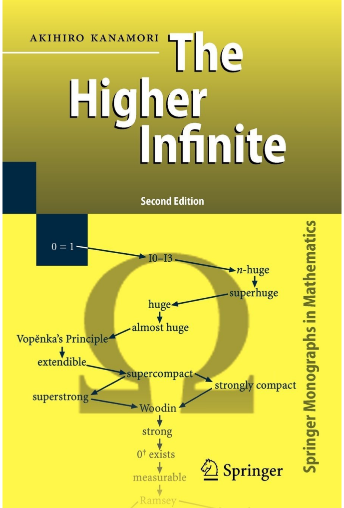
Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Course Outline

- Introduction. Formal languages and theories. Methods of proof.
- The Zermelo–Fraenkel axioms.
- The Axiom of Choice.
- Ordered sets.
- Ordinal numbers.

Birth



"*Set theory was born on that December 1873 day when Cantor established that the reals are uncountable, i.e., there is no one-to-one correspondence between the reals and the natural numbers.*" *(Kanamori [1994] 2009, p. XII)*

Naive Set Theory

Remark

Cantor's set theory is also called naive set theory.

Cantor's set definition

*"Unter einer **Menge** verstehen wir jede Zusammenfassung M von bestimmten wohlunterschiedenen Objecten m unsrer Anschauung oder unseres Denkens (welche die **Elemente** von M genannt werden) zu einem Ganzen."*
(Cantor 1895, p. 481)

*"By an **aggregate** (Menge) we are to understand any collection into a whole M of definite and separate objects m of our intuition or our thought. These objects are called the **elements** of M." (Cantor [1895] 1915, p. 85)*

*"A **set** is a collection into a whole of definite, distinct objects of our intuition or our thought. The objects are called **elements (members)** of the set."*
(Hrbacek and Jech [1978] 1999, p. 1)

Naive Set Theory

Transfinite numbers

Cantor introduced two kinds of transfinite numbers, each possessing its own arithmetic system that builds upon and expands the conventional arithmetic of natural number: Ordinal and cardinal numbers.

The Crisis in the Foundations of Mathematics

$$\text{Paradoxes} \Rightarrow \text{Crisis} \Rightarrow \begin{cases} \text{Logicism} & \text{(Russell and Whitehead)} \\ \text{Formalism} & \text{(Hilbert)} \\ \text{Intuitionism} & \text{(Brouwer)} \end{cases}$$

The Crisis in the Foundations of Mathematics

Logicism (Russell and Whitehead)

> *"The logicistic thesis is that mathematics is a branch of logic. The mathematical notions are to be defined in terms of the logical notions. The theorems of mathematics are to be proved as theorems of logic." (Kleene [1952] 1974, p. 43)*

Formalism (Hilbert)

> *"Classical mathematics shall be formulated as a formal axiomatic theory, and this theory shall be proved to be consistent, i.e., free from contradiction."*
> *(Kleene [1952] 1974, p. 53)*

The Crisis in the Foundations of Mathematics

Intuitionism (Brouwer)

> *"Intuitionism is based on the idea that mathematics is a creation of the mind. The truth of a mathematical statement can only be conceived via a mental construction that proves it to be true." (Iemhoff 2024)*

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Axiomatic set theories as a foundational system for mathematics

- "*Our axioms provide a sufficient collection of assumptions for the development of the whole of mathematics—a remarkable fact.*" *(Enderton 1977, p. 11)*

- "*But why axiomatize set theory in the first place? Well, for one thing, it is well known that set theory provides a unified framework for the whole of pure mathematics, and surely if anything deserves to be put on a sound basis it is such a foundational subject.*" *(Devlin [1979] 1993, p. 29)*

- "*Conventional mathematics is based on* ZFC *(the Zermelo–Fraenkel axioms, including the Axiom of Choice). Working withing* ZFC, *on develops: [...] All the mathematics found in basic texts on analysis, topology, algebra, etc.*" *(Kunen [2011] 2013, p. 1)*

Axiomatic Set Theories

Some axiomatic systems of set theory

- Zermelo–Fraenkel set theory (ZF)
- Zermelo–Fraenkel set theory with Choice (ZFC)
- von Neumann–Bernays–Gödel set theory (NBG)
- Morse–Kelley set theory (MK)
- Tarski–Grothendieck set theory (TG)

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But!

> *"On the other hand, we do not claim that every true fact about sets can be derived from the axioms we present. The axiomatic system is not complete in this sense." (Hrbacek and Jech [1978] 1999, p. 3)*

Foundational systems

(a) Set theories (with additional axioms)
(b) Category theories
(c) Type theories
(d) Univalent foundations
(e) Homotopy type theories

Foundational systems[†]

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[†]See, e.g., (Centrone, Kant and Sarikaya 2019).

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Logical constants

$\wedge$	(and)	conjunction
$\vee$	(or)	inclusive disjunction†
$\rightarrow$	(if __, then __)	conditional, (material) implication
$\neg$	(not)	negation
$\leftrightarrow$	(if and only if)	bi-conditional, (material) equivalence
$\forall x$	(for every x)	universal quantifier
$\exists x$	(there exists a x)	existential quantifier
$\exists! x$	(there exists one and only one x)	unique existential quantifier
$\doteq$	(equal)	equality, identity‡

†One or the other or both.

‡The textbook uses the symbol '$=$'.

Logical and Mathematical Preliminaries

Meta-logical symbol

$:=$	(is defined as)	left side is being defined by the right side
$=:$	(is defined as)	right side is being defined by the left side

About sets (p. 3)

"*A **set** X is a collection of objects called the **elements**, or **members**, of X. We sometimes use words like **family** [...] to aid comprehension.*"

"*We write $x \in X$ to express that the object x **is an element** of the set X and $y \notin X$ to say that y **is not an element** of X*".

"*Two sets X and Y are equal if and only if they contain the same elements or, equivalently, if and only if every element of X is an element of Y and vice versa.*"

Numerical sets

$\mathbb{N}$: Set of natural numbers (including the zero)
$\mathbb{Z}$: Set of all integers (positives, negatives and zero)
$\mathbb{Q}$: Set of rational numbers
$\mathbb{R}$: Set of real numbers

About the Cartesian product (p. 4)

"*We write* $X \times Y$ *for the* ***Cartesian product*** *of* X *and* Y, *i.e. the set of all ordered pairs* (x, y) *with* $x \in X$ *and* $y \in Y$*.*"

About functions (p. 4)

"*A **function** [(or **map**)] f from a set X to a set Y associates an element, $f(x)$, of Y with each element x of X. The **rule** of f, written as $x \longmapsto f(x)$, describes this process of association. The element $f(x)$ of Y is called the **image** of x under f. The **domain** of f is the set X and the **codomain** of f is the set Y.*"

The above function is denoted by

$$\begin{aligned} f : X &\longrightarrow Y \\ x &\longmapsto f(x) \end{aligned}$$

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The above function is denoted by

$$\begin{aligned} f : X &\longrightarrow Y \\ x &\longmapsto f(x) \end{aligned}$$

"*If A is a subset of the domain of this function f, the **restriction** of f to A, written as $f|_A$, is the function*"

$$\begin{aligned} f|_A : A &\longrightarrow Y \\ x &\longmapsto f(x) \end{aligned}$$

About functions (pp. 4–5)

"*The **image set** or **range** of* $f : X \longrightarrow Y$*, written as* $\mathrm{Range}(f)$*, is the set of images of* f*, namely* $\{f(x) : x \in X\}$*.*"

Definitions (informal (p. 5))

Let $f : X \longrightarrow Y$ be a function.

(a) The function f is **onto** if for each $y \in Y$ there is an $x \in X$ with $f(x) = y$.

(b) The function f is **one-one** if $f(x) = f(x')$ then $x = x'$, for all $x, x' \in X$.

(c) The function f is a **bijection**, if f is both one-one and onto.

Definition (informal (p. 5))

If $f : X \longrightarrow Y$ is a one-one function, then its **inverse function**, denoted f^{-1}, is the function defined by

$$\begin{aligned} f^{-1} : \mathrm{Range}(f) \subseteq Y &\longrightarrow X \\ y &\longmapsto \text{the unique } x \text{ such that } f(x) = x \end{aligned}$$

Definition (informal (p. 5))

If $f : X \longrightarrow Y$ and $g : Y \longrightarrow Z$ are functions then the **composite function** (or the **composition of g after f**), denoted $g \circ f$, is the function defined by

$$\begin{aligned} g \circ f : X &\longrightarrow X \\ x &\longmapsto g(f(x)) \end{aligned}$$

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$$
\begin{aligned}
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\end{aligned}
$$

Theorem

"*If f and g are both bijections with the codomain of f equal to the domain of g, then the composition $g \circ f$ is also a bijection.*" *(p. 5)*

Acronyms

AC	(Axiom of Choice)
FOL	(First-Order Logic)
FOLEQ	(First-Order Logic with Equality)
ZFC	(Zermelo–Fraenkel Set Theory with Choice)

Logical and Mathematical Preliminaries

Homework

To read Chapter 1 of the textbook.

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